Adithya Mylavarapu Naga

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About Me

Enthusiastic and Organized System and control engineer from the University of Twente, Netherlands. I utilize my interpersonal skills to promote effective teamwork, breaking down problems into accessible steps.
Hobby: Ultimate Frisbee

Experience

Software Engineer – Concr3de B.V.

- Developed an internal diagnostics engine and log-analysis libraries in C#
- Built pipelines to parse, synchronize, and reconstruct machine logs (motion, pressure, temperature, state events).
- Responsible for hardware and software integration of new camera system, currently developing an image-analysis pipeline for machine monitoring and fault investigation.
- Commissioned industrial machines: motor tuning, safety validation, updating PLC/drive configuration, and on-site debugging.

August 2025 – Present | Rotterdam | C#

Student Assistant – Advanced Software Development for Robotics

- Teaching assistant for Advanced Software Development for Robotics Course
- Hands on Experience with control of real-time mechatronic setup using RTOS and RPi

Feb – Apr 2023 | Xenomai, ROS2, C++

Robotics Intern – Aziobot B.V.

- Gained experience in robot software design and simulation of Autonomous SLAM in ROS.
- Implemented on Self-Exploration and Mapping, sensor fusion for a floor scrubber robot and optimized navigation.

Sept 2022 – Jan 2023 | ROS, Gazebo

Thesis

Master thesis - Safety Metric for Human–Aerial Robot Collaboration

- Designed an NMPC for UAVs to optimize trajectory under aerodynamic disturbances.
- Modelling, Contol and Simulink and Gazebo Simulation for safety analysis.

Jul 2023 – Jul 2024 | CasADi, Acados, MATLAB

Bachelor Thesis - Collaborative Multi-Robot System for Material Handling

- Developed an algorithm to maintain multi-robot formation in object transportation.
- Implemented an path planning algorithm for robot navigation.

Jan – Jun 2021 | Python

Control Simulations

Underactuated Offshore Crane Video

- Derived and implemented a nonlinear underactuated coupled crane model from literature.
- Designed and compared multiple controllers: PD, energy-based nonlinear control.
- Evaluated stability, swing suppression, and sea wavesdisturbance rejection in simulation.

Torque-Limited Pendulum Video

- Dynamic Programming (Value Iteration) for Swing-Up policy with actuator limits
- Formulated swing-up as an optimal control problem under torque constraints.
- Implemented value-iteration based dynamic programming to compute global policies.

Cartesian Manipulator Video

- Modeling and Control of a simplified 3D Printer.
- Multi-axis trajectory tracking and motion planning for a Cartesian system

Education

M.Sc. Systems and Control
Robotics and Mechatronics
Sept 2021 – July 2024
Universitiet Twente, Netherlands

B.Tech Mechatronics Engineering
June 2017 – July 2021
SRM Institute of Science & Technology, India

Skills

Software & Frameworks

→ Professional
- C#
- Python
- MATLAB / Simulink
- Git
- CasADi
- Acados

→ Intermediate
- C++
- ROS / ROS2
- Gazebo
- RTOS

Coursework

- Modeling & Dynamics
- Optimal Control
- Underactuated Systems
- Trajectory Optimization
- Computer Vision

Languages:

- English (C1)
- German (B1)

Contributions

Patent (2020) : On-board Hardware Addressing System for Modular Robots
Publication (2022) : Composite Robot Algorithm for Multi-Robot Formation
Award : Runner-up – Make-a-thon 4.0 : Modular Robots

Certificates - Online Courses

2020 : Self-Driving Cars Specialization – University of Toronto
2020 : Control of Mobile Robots – Georgia Tech
2020 : Autonomous Mobile Robots – ETH Zurich